CLASS TEST-1

Course Name	INTERNET PROGRAMMING
Course Code	CSA43
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1. Student Registration Validation: Write a JavaScript program to validate a student registration form containing a full name, email address, 10-digit mobile number, date of birth in DD-MM-YYYY format, and password. The password must contain at least one uppercase letter, one lowercase letter, one digit, and one special character, with a minimum length of eight characters.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Student Registration</title>
</head>
<body>
<h2>Student Registration Form</h2>
<form onsubmit="return validate()">
  Full Name: <input type="text" id="name" required><br><br>
  Email: <input type="text" id="email" required><br><br>
  Mobile: <input type="text" id="mobile" required><br><br>
  DOB: <input type="text" id="dob" placeholder="DD-MM-YYYY" required><br><br>
  Password: <input type="password" id="pass" required><br><br>
  <button type="submit">Register</button>
</form>
<p id="result"></p>
<script>
function validate() {
  let n = document.getElementById("name").value;
  let e = document.getElementById("email").value;
  let m = document.getElementById("mobile").value;
  let d = document.getElementById("dob").value;
  let p = document.getElementById("pass").value;
  let res = document.getElementById("result");

  let nr = /^[A-Za-z]+( [A-Za-z]+)+$/;
  let er = /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/;
  let mr = /^\d{10}$/;
  let dr = /^(\d{2})-(\d{2})-(\d{4})$/;
  let pr = /^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[^A-Za-z0-9]).{8,}$/;

  let errors = [];
  if (!nr.test(n)) errors.push("Invalid name (enter first and last name, letters only)");
  if (!er.test(e)) errors.push("Invalid email address");
  if (!mr.test(m)) errors.push("Mobile number must contain exactly 10 digits");

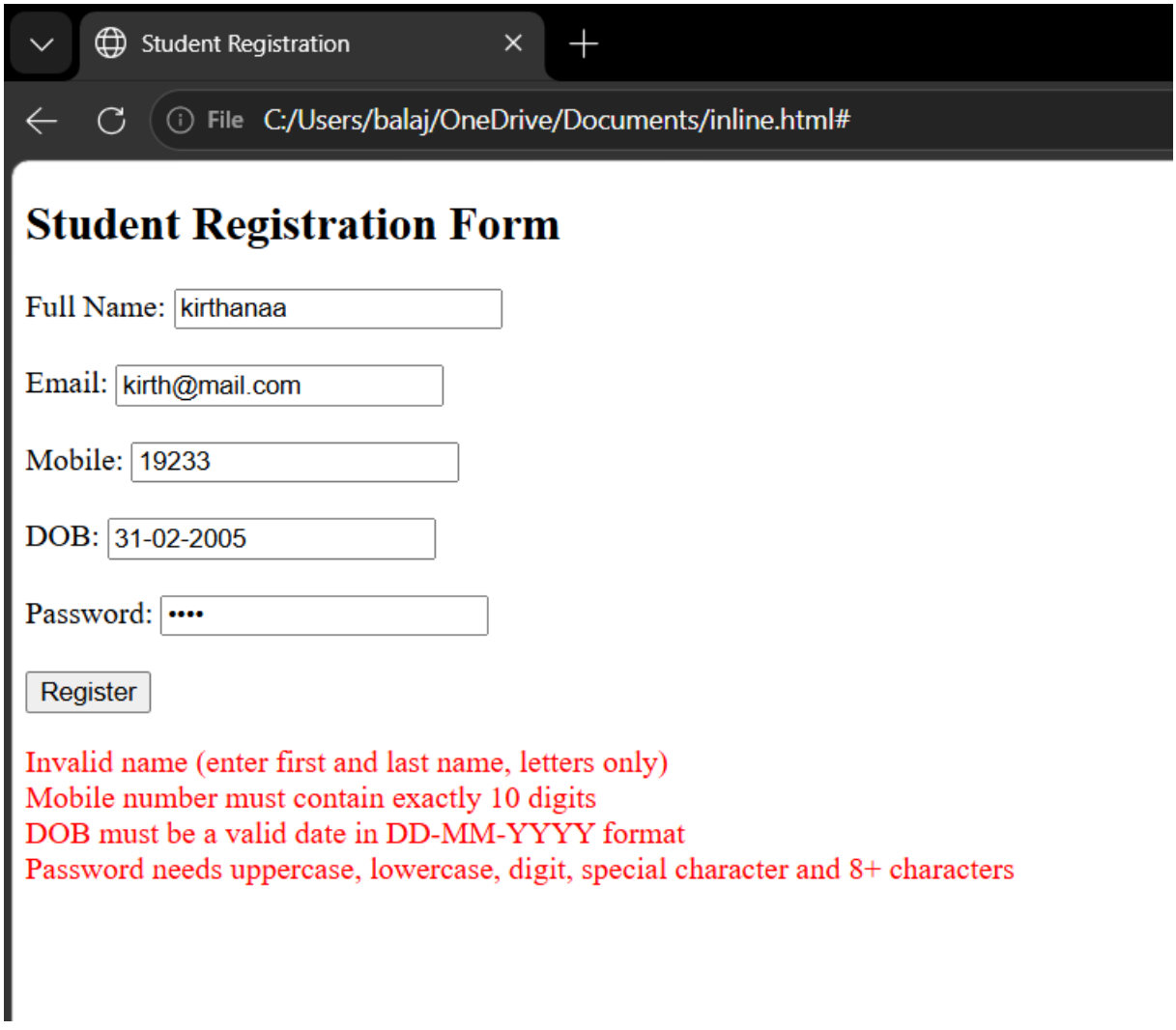
  let dm = dr.exec(d);
```

```
  let dateOk = false;
  if (dm) {
    let dt = new Date(dm[3], dm[2] - 1, dm[1]);
    dateOk = dt.getDate() == dm[1] && dt.getMonth() == dm[2] - 1;
  }
  if (!dateOk) errors.push("DOB must be a valid date in DD-MM-YYYY format");
  if (!pr.test(p)) errors.push("Password needs uppercase, lowercase, digit, special character and
8+ characters");

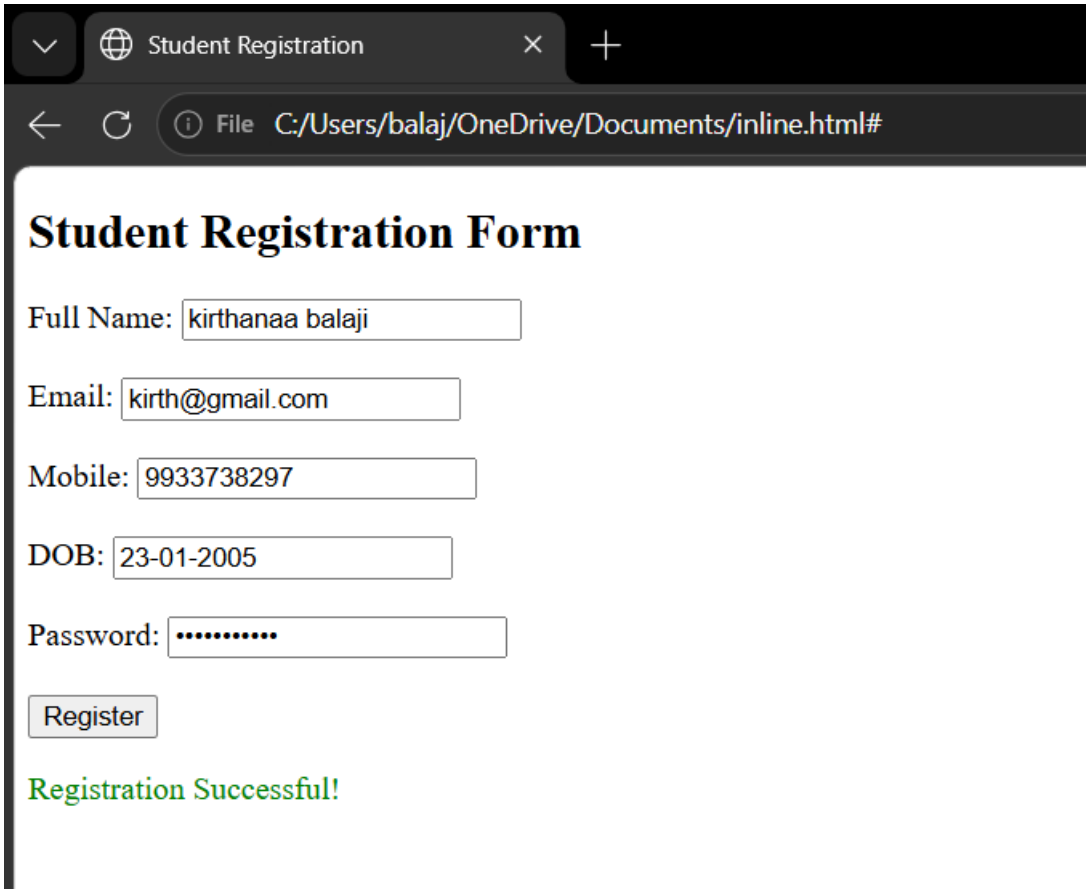
  if (errors.length) {
    res.style.color = "red";
    res.innerHTML = errors.join("<br>");
  } else {
    res.style.color = "green";
    res.innerHTML = "Registration Successful!";
  }
  return false;
}
</script>
</body>
</html>
```

OUTPUT:

Invalid input



Valid input



2. Password Strength Validation: Write a JavaScript program to validate a password with the following rules: minimum 10 characters, at least two digits, one uppercase letter, one lowercase letter, one special character, and no whitespace. Display an appropriate error message for each failed condition.

CODE:

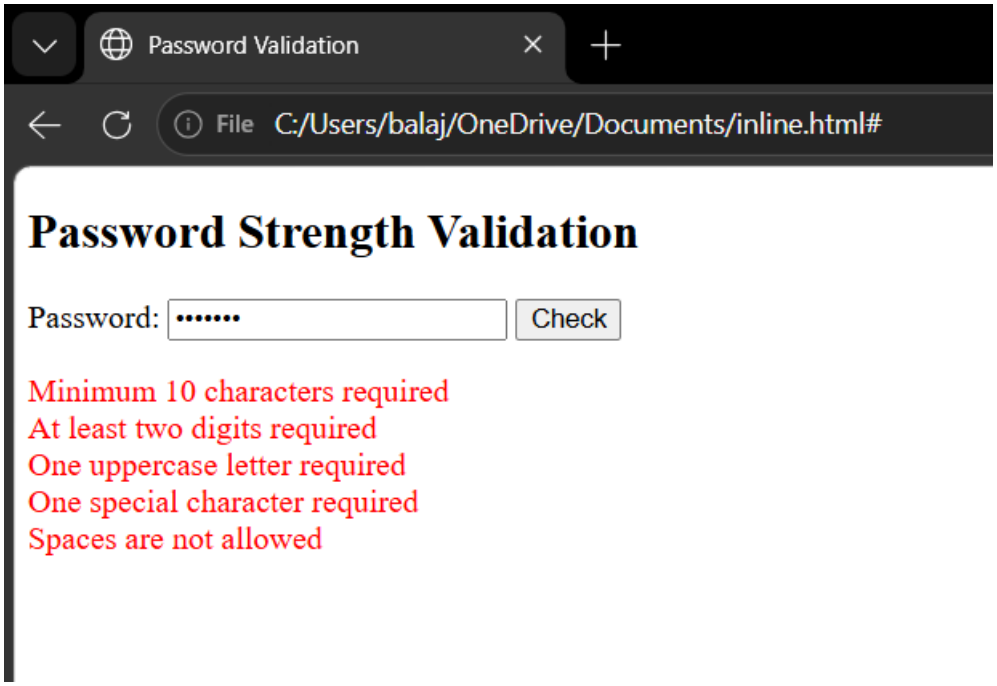
```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Password Validation</title>
</head>
<body>
<h2>Password Strength Validation</h2>
<form onsubmit="return validate()">
  Password: <input type="password" id="pass" required>
  <button type="submit">Check</button>
</form>
<p id="result"></p>
<script>
function validate() {
  let p = document.getElementById("pass").value;
  let errors = [];
  if (p.length < 10)
    errors.push("Minimum 10 characters required");
  if ((p.match(/\d/g) || []).length < 2)
    errors.push("At least two digits required");
  if (!/[A-Z]/.test(p))
    errors.push("One uppercase letter required");
  if (!/[a-z]/.test(p))
```

```
    errors.push("One lowercase letter required");
  if (!/[^A-Za-z0-9\s]/.test(p))
    errors.push("One special character required");
  if (/\s/.test(p))
    errors.push("Spaces are not allowed");

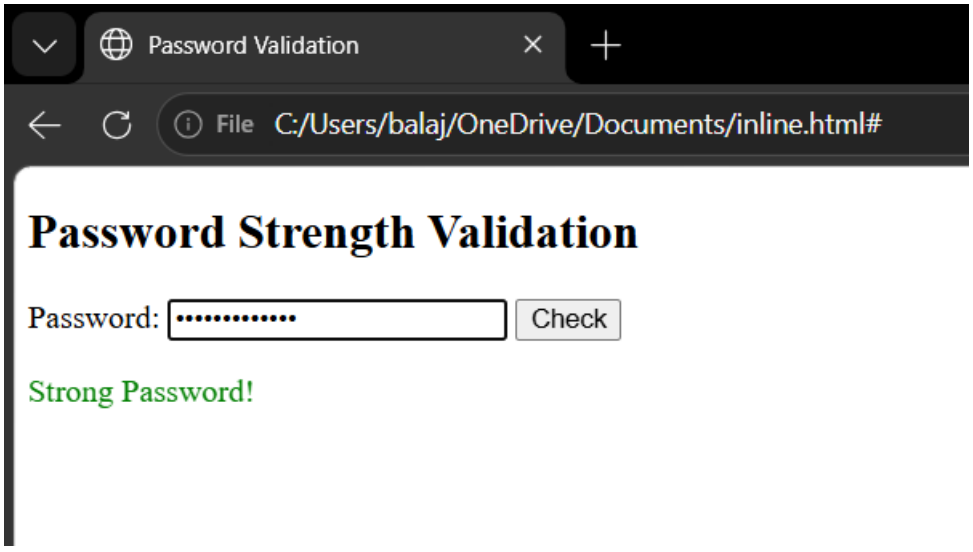
  let res = document.getElementById("result");
  res.style.color = errors.length ? "red" : "green";
  res.innerHTML = errors.length ? errors.join("<br>") : "Strong Password!";
  return false;
}
</script>
</body>
</html>
```

OUTPUT:

Weak password



Strong password



3. Email Address Validation: Write a JavaScript program to validate email addresses according to these rules: the address must contain exactly one @, a non-empty username, a domain name, and a valid-looking extension of at least two letters. Reject spaces and consecutive dots.

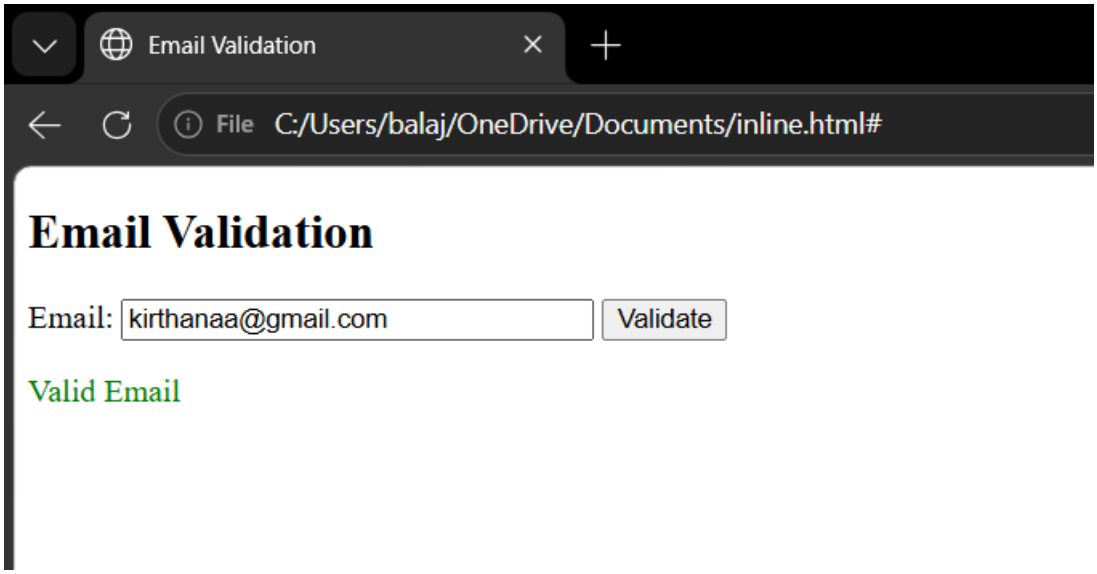
CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Email Validation</title>
</head>
<body>
<h2>Email Validation</h2>
<form onsubmit="return validate()">
  Email: <input type="text" id="email" size="30" required>
  <button type="submit">Validate</button>
</form>
<p id="result"></p>
<script>
function validate() {
  let e = document.getElementById("email").value;
  let at = (e.match(/@/g) || []).length;
  let pattern =
    /^[A-Za-z0-9_%+-]+(\.[A-Za-z0-9_%+-]+)*@[A-Za-z0-9-]+(\.[A-Za-z0-9-]+)*\.[A-Za-
z]{2,}$/;
  let errors = [];
  if (/\s/.test(e)) errors.push("Spaces are not allowed");
  if (at !== 1) errors.push("Email must contain exactly one @");
  if (/\.\./.test(e)) errors.push("Consecutive dots are not allowed");
  if (errors.length === 0 && !pattern.test(e))
    errors.push("Check the username, domain name and extension (minimum two
letters)");

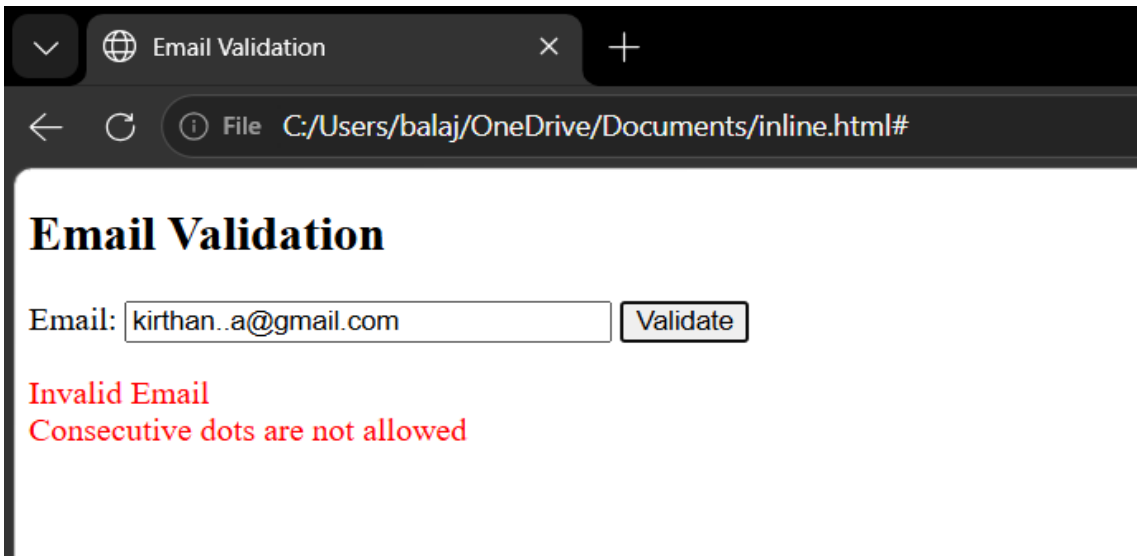
  let res = document.getElementById("result");
  if (errors.length) {
    res.style.color = "red";
    res.innerHTML = "Invalid Email<br>" + errors.join("<br>");
  } else {
    res.style.color = "green";
    res.innerHTML = "Valid Email";
  }
  return false;
}
</script>
</body>
</html>
```

OUTPUT:

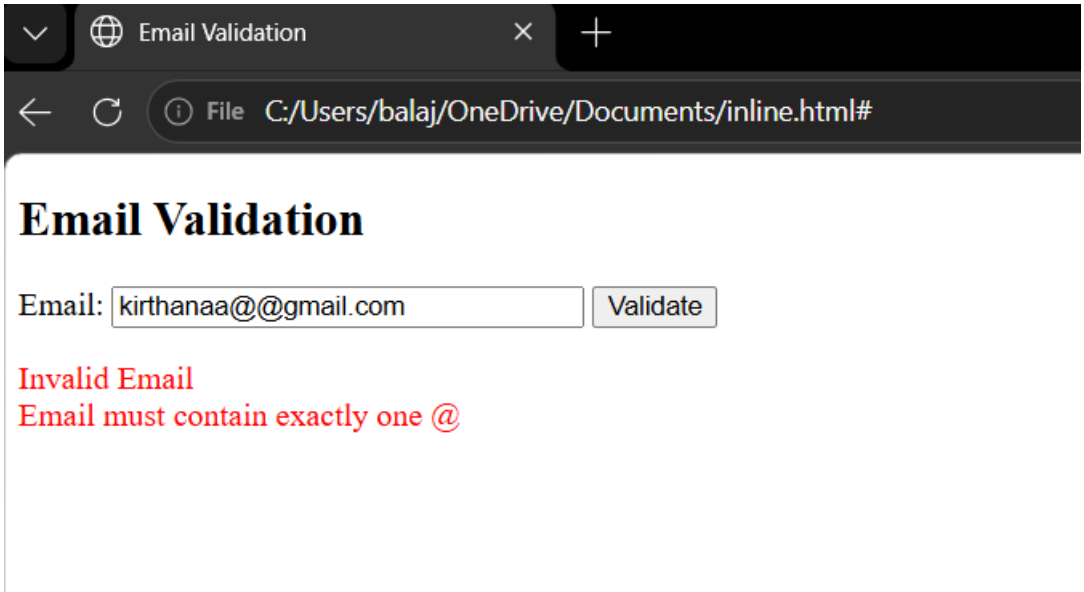
Valid email



Invalid email (consecutive dots)



Invalid email (two @ symbols)



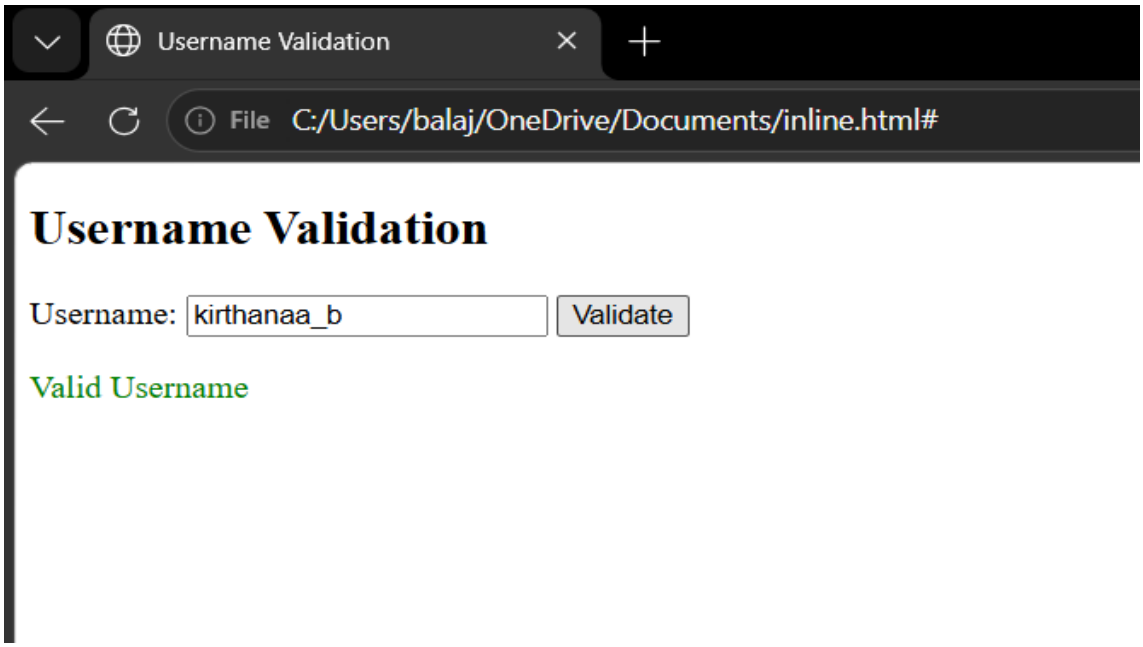
4. Username Validation: Write a JavaScript program to validate a username that must start with a letter, contain only letters, digits, and underscores, have a length of 6–15 characters, and must not end with an underscore or contain consecutive underscores.

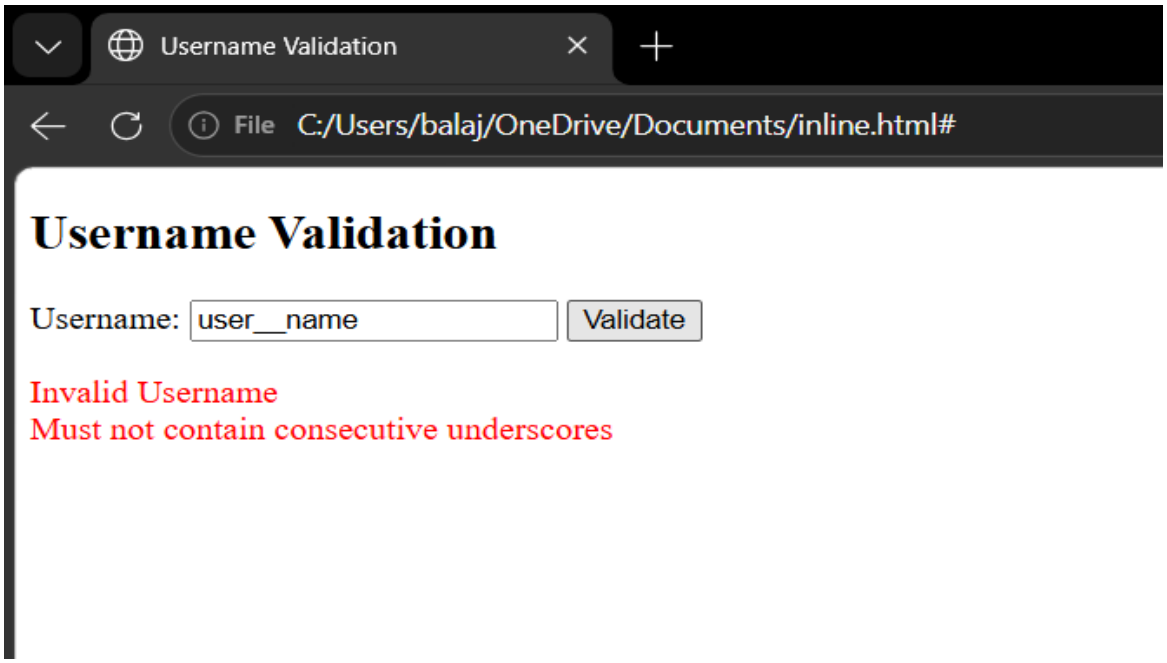
CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Username Validation</title>
</head>
<body>
<h2>Username Validation</h2>
<form onsubmit="return validate()">
  Username: <input type="text" id="user" required>
  <button type="submit">Validate</button>
</form>
<p id="result"></p>
<script>
function validate() {
  let u = document.getElementById("user").value;
  let pattern = /^(?=.{6,15}$)(?!.*__)[A-Za-z][A-Za-z0-9_]*[A-Za-z0-9]$/;
  let errors = [];
  if (!/^[A-Za-z]/.test(u)) errors.push("Must start with a letter");
  if (/[^A-Za-z0-9_]/.test(u)) errors.push("Only letters, digits and underscores allowed");
  if (u.length < 6 || u.length > 15) errors.push("Length must be 6 to 15 characters");
  if (/_$/.test(u)) errors.push("Must not end with an underscore");
  if (/__/.test(u)) errors.push("Must not contain consecutive underscores");

  let res = document.getElementById("result");
  if (pattern.test(u)) {
    res.style.color = "green";
    res.innerHTML = "Valid Username";
  } else {
    res.style.color = "red";
    res.innerHTML = "Invalid Username<br>" + errors.join("<br>");
  }
  return false;
}
</script>
</body>
</html>
```

OUTPUT:





5. Extract and Analyse Data: Write a JavaScript program to extract all mobile numbers, email addresses, and dates from a paragraph using regular expressions. Display each category separately.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Extract Data</title>
</head>
<body>
<h2>Extract Data Using Regular Expressions</h2>
<textarea id="text" rows="5" cols="55">Contact Kirthanaa at 9876543210 or 8123456780. Email kirthanaa@example.com and admin@simats.ac.in. Important dates: 12-10-2026 and 05/11/2026. Alternate number 7012345678.</textarea>
<br><br>
```

```
<button onclick="extract()">Extract Data</button>
<h3>Mobile Numbers</h3>
<p id="mobile"></p>
<h3>Email Addresses</h3>
<p id="email"></p>
<h3>Dates</h3>
<p id="date"></p>
<script>
function extract() {
  let t = document.getElementById("text").value;
  let m = t.match(/\b[6-9]\d{9}\b/g) || [];
  let e = t.match(/\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}\b/g) || [];
  let d = t.match(/\b(0[1-9]|[12]\d|3[01])[-\/](0[1-9]|1[0-2])[-\/]\d{4}\b/g) || [];
  document.getElementById("mobile").innerHTML = m.join("<br>") || "None found";
  document.getElementById("email").innerHTML = e.join("<br>") || "None found";
  document.getElementById("date").innerHTML = d.join("<br>") || "None found";
}
</script>
</body>
</html>
```

OUTPUT:

